Data Budget

DA VINCI SATELLITE

TT&C and GS&O

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Introduction

This report presents the data budget for a 2U CubeSat mission using a VHF uplink and a UHF downlink. The objective is to quantify how much user data can be reliably transferred between the satellite and the ground station during nominal operations, given the constraints imposed by the communication hardware, the AX.25 protocol stack, and the available contact time.

The satellite is equipped with an ISIS TRXVU transceiver, which provides a UHF downlink in the 430–450 MHz band and a VHF uplink in the 140–150 MHz band. The downlink supports BPSK modulation with data rates from 1200 to 9600 bit/s, while the uplink uses FSK at 9600 bit/s. Both links use the AX.25 protocol with UI frames, and the maximum frame payload is 235 bytes for the downlink and 200 bytes for the uplink. The antenna system is an ISIS AntS deployable antenna, which can be configured as a turnstile, dipole, or monopole arrangement.

The data budget is calculated from the physical data rate, the frame structure, the protocol overhead, and the expected contact time per pass. The frame structure and protocol overhead are derived from the mission's transmission documentation, while the contact time is treated as a parameter that depends on the orbit and the ground station network. The results are presented for several downlink scenarios, including pure telemetry, payload data, and a mixed case, together with a sensitivity analysis on contact duration and the number of passes per day.

The scope of this report is limited to the data budget. The link budget, which determines whether the RF link can be established with sufficient margin, is not included here. The data budget results therefore assume that the link is available and that the physical data rate can be sustained during the contact window. This separation is intentional: the link budget is addressed separately, and the present analysis focuses on how much data can be transferred once the link is operational.

The remainder of this report is organized as follows. Section 1 describes the methodology, including the input parameters, the frame structure, and the calculation approach. Section 2 presents the results for the different downlink scenarios. Section 3 discusses the assumptions, limitations, and implications of the results. References and additional data are provided at the end of the report.

1 Methodology

1.1 Overview

The data budget is computed from three quantities: the physical data rate of the downlink, the fraction of each frame that carries useful payload data, and the total time available for downlink during a contact pass. The first two are determined by the communication hardware and the AX.25 protocol stack. The third depends on the orbit and the ground station network, and is treated here as a parameter.

The calculation is performed separately for each downlink data type, because different frame structures carry different amounts of overhead. The results are then combined into an aggregate data budget for a representative operational scenario.

1.2 Input Parameters

The input parameters used in this analysis are listed in Table 1. They are grouped into three categories: communication hardware, frame structure, and contact parameters.

Table 1: Input parameters for the data budget.

Parameter	Value	Source
Communication hardware		
Downlink frequency band	430–450 MHz	TRXVU ICD
Uplink frequency band	140–150 MHz	TRXVU ICD
Downlink modulation	BPSK	TRXVU ICD
Uplink modulation	FSK	TRXVU ICD
Downlink data rate	9600 bit/s	TRXVU ICD
Uplink data rate	9600 bit/s	TRXVU ICD
Downlink maximum frame payload	235 bytes	TRXVU ICD
Uplink maximum frame payload	200 bytes	TRXVU ICD
Protocol	AX.25 (UI frames)	TRXVU ICD
Encoding	NRZ-I, no FEC	TRXVU ICD
Frame structure		
Downlink common header	5 bytes	Frame Structure v2.0
Uplink common header	7 bytes	Frame Structure v2.0
AX.25 metadata	18 bytes per frame	Frame Structure v2.0
Picture chunk data	64 bytes per frame	Transmission Manual
Picture chunk header	6 bytes per frame	Transmission Manual
Telemetry preamble	7 bytes per frame	Transmission Manual
Contact parameters		
Orbit altitude	500 km (assumed)	–
Minimum elevation	10° (assumed)	–
Contact duration per pass	10 min (assumed)	–
Passes per day	3 (assumed)	–
Effective communication fraction	80% (assumed)	–

The values marked as assumed are typical for a 2U CubeSat in low Earth orbit with a single ground station. They are used here to produce a baseline data budget. Once the actual orbit and ground station network are known, these values can be replaced without changing the calculation method.

1.3 Frame Structure and Protocol Overhead

1.3.1 Downlink Frame

Each downlink frame consists of the AX.25 metadata, the common frame header, and the payload data. The AX.25 metadata is 18 bytes and is present on every frame. The common frame header is 5 bytes and contains the command message counter, the frame length, the subsystem identifier, and the frame type. The remaining bytes carry the payload data, up to a maximum of 230 bytes.
The effective payload fraction of a downlink frame is therefore

$$\eta_{\text{DL}} = \frac{L_{\text{payload}}}{L_{\text{payload}} + L_{\text{header}} + L_{\text{AX.25}}}, \tag{1}$$

where L_{payload} is the payload size in bytes, $L_{\text{header}} = 5$ bytes, and $L_{\text{AX.25}} = 18$ bytes. For a full-size frame with $L_{\text{payload}} = 230$ bytes, the efficiency is

$$\eta_{\text{DL,full}} = \frac{230}{230 + 5 + 18} = \frac{230}{253} \approx 90.9\%. \tag{2}$$

For smaller frames, such as telemetry or picture chunks, the efficiency is lower because the fixed overhead is amortized over fewer payload bytes. The efficiency is therefore computed separately for each data type.

1.3.2 Picture Chunk Frame

A picture downlink chunk carries 64 bytes of picture data. The frame structure is a common header of 5 bytes, followed by one byte for the picture identifier, followed by the 64 bytes of picture data. Including the AX.25 metadata, the total transmitted size is

$$L_{\text{chunk}} = 18 + 5 + 1 + 64 = 88 \,\text{bytes}, \tag{3}$$

so the payload efficiency of a picture chunk is

$$\eta_{\text{chunk}} = \frac{64}{88} \approx 72.7\%. \tag{4}$$

1.3.3 Telemetry Frame

Telemetry frames have a variable payload size depending on the telemetry type. Each telemetry frame begins with a preamble of 7 bytes, which includes the command code, content code, telemetry code, and a four-byte timestamp. The payload that follows depends on the telemetry data set, as described in the transmission manual. Because the payload size varies, the efficiency is computed per telemetry type using the same method as above.

1.3.4 Uplink Frame

Although the data budget is primarily concerned with the downlink, the uplink frame structure is included for completeness. An uplink frame consists of the AX.25 metadata (18 bytes), the common header (7 bytes), and the payload data, up to a maximum of 193 bytes. The effective payload fraction is computed in the same way as for the downlink.

1.4 Data Budget Calculation

The data budget is calculated in three steps: the effective data rate, the data volume per pass, and the daily data volume.

1.4.1 Effective Data Rate

The effective data rate is the physical data rate multiplied by the payload efficiency:

$$R_{\text{eff}} = R_{\text{phys}} \cdot \eta, \tag{5}$$

where $R_{\text{phys}} = 9600$ bit/s and η is the payload efficiency for the data type under consideration. For a picture chunk, for example,

$$R_{\text{eff,chunk}} = 9600 \cdot 0.727 \approx 6980 \,\text{bit/s} \approx 872 \,\text{bytes/s}. \tag{6}$$

1.4.2 Data Volume per Pass

The data volume per pass is the effective data rate multiplied by the effective contact time:

$$V_{\text{pass}} = R_{\text{eff}} \cdot T_{\text{pass}} \cdot f_{\text{eff}}, \tag{7}$$

where T_{pass} is the geometric contact duration and f_{eff} is the fraction of that duration available for useful downlink after accounting for tracking, synchronization, and protocol handshaking. Using the assumed values of $T_{\text{pass}} = 10$ min and $f_{\text{eff}} = 0.8$,

$$V_{\text{pass,chunk}} = 872 \cdot 600 \cdot 0.8 \approx 418{,}560 \,\text{bytes} \approx 409 \,\text{KB}. \tag{8}$$

1.4.3 Daily Data Volume

The daily data volume is the data volume per pass multiplied by the number of passes per day:

$$V_{\text{day}} = V_{\text{pass}} \cdot N_{\text{pass}}, \tag{9}$$

where N_{pass} is the number of effective passes per day. Using the assumed value of $N_{\text{pass}} = 3$,

$$V_{\text{day,chunk}} = 409 \cdot 3 \approx 1227 \,\text{KB} \approx 1.2 \,\text{MB}. \tag{10}$$

1.5 Scenarios

Three downlink scenarios are considered:

1. **Telemetry only.** All downlink frames carry telemetry data. The payload efficiency is computed from the telemetry frame structure.
2. **Payload data only.** All downlink frames carry picture chunks. The payload efficiency is $\eta_{\text{chunk}} \approx 72.7\%$.
3. **Mixed scenario.** The downlink is shared between telemetry and payload data in a representative proportion. The aggregate data volume is the weighted sum of the contributions from each data type.

1.6 Assumptions and Limitations

The following assumptions are made in this analysis:

- The physical data rate is sustained for the entire contact window.
- The link is established with sufficient margin, as determined by a separate link budget analysis that is outside the scope of this report.
- The frame error rate is low enough that retransmissions do not significantly reduce the effective data rate. The packet error rate for a 235-byte frame at a bit error rate of 10^{-5} is approximately 1.6%, as computed from the relation $\text{PER} = 1 - (1 - \text{BER})^{L}$.
- The contact duration, number of passes, and effective communication fraction are representative values for a 2U CubeSat in low Earth orbit. They are not derived from a specific orbit or ground station network.
- Protocol overhead is computed from the frame structure documentation and does not include additional overhead from the ground station or the network layer.

The main limitation of this analysis is that the contact parameters are assumed rather than derived. The data budget is therefore a baseline that can be refined once the orbit and ground station network are defined. The calculation method, however, remains valid and can be applied directly to the actual parameters.

2 Results

2.1 Overview

This section presents the data budget for the 2U CubeSat mission. The results are organized into three scenarios: telemetry-only downlink, payload-only downlink, and a mixed scenario representative of nominal operations. For each scenario, the effective data rate, the data volume per pass, and the daily data volume are reported. A sensitivity analysis follows, showing how the results depend on contact duration and the number of passes per day.

All results are based on the input parameters listed in Table 1. The assumed contact parameters are an orbit altitude of 500 km, a minimum elevation of 10°, a contact duration of 10 min per pass, three passes per day, and an effective communication fraction of 80%. These values are representative of a 2U CubeSat in low Earth orbit with a single ground station and can be replaced once the actual orbit and ground station network are defined.

2.2 Frame Efficiency

Before presenting the scenario results, it is useful to summarize the payload efficiency of the different frame types. The efficiency is defined as the ratio of useful payload bytes to the total transmitted bytes, including the AX.25 metadata and the common frame header.

Table 2: Payload efficiency of different downlink frame types.

Frame type	Payload (bytes)	Total (bytes)	Efficiency
Picture chunk	64	88	72.7%
Full-size data frame	230	253	90.9%
Short telemetry (OBC)	14	44	31.8%
Medium telemetry (EPS)	30	60	50.0%
Transmitter telemetry	23	53	43.4%
Receiver telemetry	14	44	31.8%

The picture chunk frame has a high efficiency because it carries 64 bytes of payload data. The full-size data frame has the highest efficiency because it uses the maximum payload size of 230 bytes. Telemetry frames have lower efficiency because their payload is small and the fixed overhead is amortized over fewer bytes.

2.3 Scenario 1: Telemetry-Only Downlink

In this scenario, all downlink frames carry telemetry data. The payload efficiency depends on the telemetry type. For a conservative estimate, a medium telemetry frame with a payload of 30 bytes is used, giving an efficiency of 50.0%.

2.3.1 Effective Data Rate

$$R_{\text{eff}} = R_{\text{phys}} \cdot \eta = 9600 \cdot 0.50 = 4800\,\text{bit/s} = 600\,\text{bytes/s}. \tag{11}$$

2.3.2 Data Volume per Pass

$$V_{\text{pass}} = R_{\text{eff}} \cdot T_{\text{pass}} \cdot f_{\text{eff}} = 600 \cdot 600 \cdot 0.8 = 288{,}000\,\text{bytes} \approx 281\,\text{KB}. \tag{12}$$

2.3.3 Daily Data Volume

$$V_{\text{day}} = V_{\text{pass}} \cdot N_{\text{pass}} = 281 \cdot 3 = 843\,\text{KB} \approx 0.84\,\text{MB}. \tag{13}$$

This scenario represents the minimum useful downlink, where the satellite only reports its health and status.

2.4 Scenario 2: Payload-Only Downlink

In this scenario, all downlink frames carry picture chunks. The payload efficiency is 72.7%.

2.4.1 Effective Data Rate

$$R_{\text{eff}} = 9600 \cdot 0.727 = 6979\,\text{bit/s} \approx 872\,\text{bytes/s}. \tag{14}$$

2.4.2 Data Volume per Pass

$$V_{\text{pass}} = 872 \cdot 600 \cdot 0.8 = 418{,}560\,\text{bytes} \approx 409\,\text{KB}. \tag{15}$$

2.4.3 Daily Data Volume

$$V_{\text{day}} = 409 \cdot 3 = 1227\,\text{KB} \approx 1.20\,\text{MB}. \tag{16}$$

This scenario represents the maximum payload downlink, assuming the satellite has enough stored pictures to fill the entire contact window.

2.5 Scenario 3: Mixed Downlink

In nominal operations, the downlink is shared between telemetry and payload data. A representative allocation is 25% telemetry and 75% payload data by time. The aggregate efficiency is the weighted average of the two efficiencies:

$$\eta_{\text{mixed}} = 0.25 \cdot 0.50 + 0.75 \cdot 0.727 = 0.125 + 0.545 = 0.670. \tag{17}$$

2.5.1 Effective Data Rate

$$R_{\text{eff}} = 9600 \cdot 0.670 = 6432\,\text{bit/s} = 804\,\text{bytes/s}. \tag{18}$$

2.5.2 Data Volume per Pass

$$V_{\text{pass}} = 804 \cdot 600 \cdot 0.8 = 385{,}920\,\text{bytes} \approx 377\,\text{KB}. \tag{19}$$

2.5.3 Daily Data Volume

$$V_{\text{day}} = 377 \cdot 3 = 1131\,\text{KB} \approx 1.10\,\text{MB}. \tag{20}$$

This scenario is the most representative of nominal operations, where the satellite reports its health and downlinks payload data in the same pass.

2.6 Summary of Results

Table 3 summarizes the results for the three scenarios.

Table 3: Summary of data budget results for the three scenarios.

Scenario	Effective rate (bytes/s)	Per pass (KB)	Per day (MB)
Telemetry only	600	281	0.84
Payload only	872	409	1.20
Mixed (25/75)	804	377	1.10

The payload-only scenario gives the highest data volume, as expected, because picture chunks have the highest payload efficiency. The telemetry-only scenario gives the lowest data volume, because telemetry frames are short and the fixed overhead is large. The mixed scenario lies between the two and is the most representative of nominal operations.

2.7 Sensitivity Analysis

The results depend on the assumed contact parameters. This section shows how the daily data volume varies with contact duration and the number of passes per day.

2.7.1 Effect of Contact Duration

Table 4 shows the daily data volume for the mixed scenario as a function of contact duration, assuming three passes per day and an effective communication fraction of 80%.

Table 4: Effect of contact duration on daily data volume (mixed scenario).

Contact duration (min)	Daily data volume (MB)
6	0.66
8	0.88
10	1.10
12	1.32
14	1.54

The daily data volume scales linearly with contact duration. A 2 min increase in contact duration adds approximately 0.22 MB per day in the mixed scenario.

2.7.2 Effect of Number of Passes

Table 5 shows the daily data volume for the mixed scenario as a function of the number of passes per day, assuming a contact duration of 10 min and an effective communication fraction of 80%.

Table 5: Effect of number of passes on daily data volume (mixed scenario).

Passes per day	Daily data volume (MB)
1	0.37
2	0.74
3	1.10
4	1.47
5	1.84

The daily data volume scales linearly with the number of passes. Each additional pass adds approximately 0.37 MB per day in the mixed scenario.

2.7.3 Combined Sensitivity

Figure 1 shows the combined effect of contact duration and number of passes on the daily data volume. The data volume increases linearly with both parameters, and the contours are equally spaced.

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Figure placeholder: contour plot of daily data volume as a function of contact duration and number of passes.

Figure 1: Combined sensitivity of daily data volume to contact duration and number of passes.

2.8 Key Observations

The following observations can be drawn from the results:

1. The payload-only scenario gives the highest data volume, approximately 1.20 MB per day under the assumed contact parameters.
2. The telemetry-only scenario gives the lowest data volume, approximately 0.84 MB per day.
3. The mixed scenario, which is the most representative of nominal operations, gives approximately 1.10 MB per day.
4. The daily data volume scales linearly with both contact duration and the number of passes per day.
5. The results are sensitive to the assumed contact parameters. A change of 2 min in contact duration or one additional pass per day changes the daily data volume by approximately 0.22 MB and 0.37 MB respectively in the mixed scenario.
6. The frame efficiency varies significantly between data types, from 31.8% for short telemetry frames to 72.7% for picture chunks. This difference should be taken into account when planning the downlink strategy.

3 Discussion

3.1 Interpretation of Results

The results show that the daily data volume for the 2U CubeSat ranges from approximately 0.84 MB for a telemetry-only downlink to approximately 1.20 MB for a payload-only downlink, with the mixed scenario giving approximately 1.10 MB per day. These values are consistent with typical 2U CubeSat missions using VHF uplink and UHF downlink, where the physical data rate is limited to 9600 bit/s and the contact time per pass is on the order of ten minutes.

The difference between the telemetry-only and payload-only scenarios is significant: the payload-only scenario delivers approximately 43% more data per day than the telemetry-only scenario. This difference is entirely due to the payload efficiency of the frame types. Picture chunks carry 64 bytes of payload per frame, while short telemetry frames may carry as little as 10–15 bytes. The fixed overhead, which includes the AX.25 metadata and the common frame header, is amortized over fewer bytes in the telemetry case, reducing the effective data rate.

This observation has a direct operational implication: if the mission requires a large volume of payload data, the downlink strategy should prioritize payload frames over telemetry frames wherever possible. Telemetry should be kept to the minimum required for monitoring satellite health, and payload data should fill the remainder of the contact window.

3.2 Effect of Frame Efficiency

The frame efficiency varies significantly between data types, from 31.8% for short telemetry frames to 72.7% for picture chunks. This variation is not a property of the communication hardware but of the data itself. The TRXVU transceiver always transmits at 9600 bit/s, but the fraction of that rate that carries useful data depends on how much payload each frame contains.

The full-size data frame, with a payload of 230 bytes, achieves an efficiency of 90.9%. This is the theoretical maximum for the current frame structure. In practice, this efficiency is only reached when the downlink carries large data blocks, such as code updates or large files. For smaller data types, the efficiency is lower.

This suggests that the mission could benefit from aggregating small data items into larger frames where possible. For example, multiple telemetry packets could be combined into a single frame, reducing the number of frames and increasing the average payload size. This would improve the overall efficiency of the downlink without changing the hardware or the physical data rate.

3.3 Sensitivity to Contact Parameters

The sensitivity analysis shows that the daily data volume scales linearly with both contact duration and the number of passes per day. A 2 min increase in contact duration adds approximately 0.22 MB per day in the mixed scenario, while an additional pass per day adds approximately 0.37 MB per day. These are significant changes relative to the baseline of 1.10 MB per day.

This linear dependence means that the data budget is highly sensitive to the ground station network. A mission with access to multiple ground stations, or with a ground station at a favorable latitude, could achieve a substantially higher daily data volume than a mission with a single ground station. Conversely, a mission with limited ground station access would need to prioritize its downlink data more carefully.

It should be noted that the contact parameters used in this analysis are assumed values, not derived from a specific orbit or ground station network. The actual contact duration and number of passes will depend on the orbit altitude, the minimum elevation angle, and the geographic distribution of the ground stations. Once these parameters are known, the data budget can be refined using the same calculation method.

3.4 Relationship to Link Budget

This report is limited to the data budget. The link budget, which determines whether the RF link can be established with sufficient margin, is not included here. The data budget results therefore assume that the link is available and that the physical data rate of 9600 bit/s can be sustained during the contact window.

This assumption is reasonable for a 2U CubeSat with an ISIS TRXVU transceiver and an ISIS AntS antenna, provided that the ground station has sufficient gain and that the satellite is above the minimum elevation angle. However, the link budget should be verified separately to ensure that the required margin is available, particularly at low elevation angles where the path loss is highest.

If the link budget shows that the margin is insufficient at 9600 bit/s, the data rate could be reduced to 4800 bit/s or 2400 bit/s. This would reduce the data volume proportionally but would also reduce the required signal-to-noise ratio, potentially allowing the link to close. The data budget calculation method presented in this report remains valid for any data rate; only the physical data rate parameter would need to be updated.

3.5 Assumptions and Limitations

The main assumptions of this analysis are as follows:

- The physical data rate is sustained for the entire contact window.
- The link is established with sufficient margin, as determined by a separate link budget analysis.

- The frame error rate is low enough that retransmissions do not significantly reduce the effective data rate. The packet error rate for a 235-byte frame at a bit error rate of 10^{-5} is approximately 1.6%, which is small but not negligible.
- The contact duration, number of passes, and effective communication fraction are representative values for a 2U CubeSat in low Earth orbit.
- Protocol overhead is computed from the frame structure documentation and does not include additional overhead from the ground station or the network layer.

The main limitation of this analysis is that the contact parameters are assumed rather than derived. The data budget is therefore a baseline that can be refined once the orbit and ground station network are defined. The calculation method, however, remains valid and can be applied directly to the actual parameters.

A second limitation is that the payload efficiency values are based on the frame structure documentation and assume that frames are filled to their maximum payload size. In practice, some frames may be partially filled, which would reduce the efficiency. This effect is expected to be small for picture chunks, which are always 64 bytes, but could be more significant for telemetry frames, which have variable payload sizes.

3.6 Recommendations

Based on the results and discussion, the following recommendations are made:

1. **Prioritize payload data.** The payload-only scenario delivers approximately 43% more data per day than the telemetry-only scenario. The downlink strategy should prioritize payload frames over telemetry frames wherever possible.
2. **Minimize telemetry overhead.** Telemetry should be kept to the minimum required for monitoring satellite health. Where possible, multiple telemetry packets should be aggregated into a single frame to improve the average payload size.
3. **Verify the link budget.** The data budget assumes that the link is available at 9600 bit/s. The link budget should be verified separately to ensure that the required margin is available, particularly at low elevation angles.
4. **Refine the contact parameters.** The contact parameters used in this analysis are assumed values. Once the orbit and ground station network are defined, the data budget should be recalculated using the actual values.
5. **Consider the ground station network.** The daily data volume scales linearly with the number of passes per day. A mission with access to multiple ground stations could achieve a substantially higher daily data volume than a mission with a single ground station.

3.7 Conclusion

This report has presented the data budget for a 2U CubeSat mission using a VHF uplink and a UHF downlink. The daily data volume ranges from approximately 0.84 MB for a telemetry-only downlink to approximately 1.20 MB for a payload-only downlink, with a mixed scenario giving approximately 1.10 MB per day under the assumed contact parameters. The results are sensitive to the contact duration and the number of passes per day, and the frame efficiency varies significantly between data types.

The data budget is a baseline that can be refined once the orbit and ground station network are defined. The calculation method presented in this report remains valid for any set of parameters and can be applied directly to the actual mission configuration. The link budget, which is outside the scope of this report, should be verified separately to ensure that the required margin is available at the chosen data rate.

References

[1] ISIS – Innovative Solutions In Space, *Transceiver ICD v1.5 Rev D*, ISIS.TRXVU.ICD.001, 2019.

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[3] G. Pieroni and D. Perez Trias, *Frame Design Manual*, Frame_Structure_07-26_v2-0, July 2026.

[4] *Transmission Manual*, Internal mission documentation, 2026.

[5] T. L. Fox, *AX.25 Amateur Packet-Radio Link-Layer Protocol, Version 2.2*, American Radio Relay League, 1997.

[6] Packet Error Rate calculation, PER $= 1 - (1 - \text{BER})^L$, based on the TRXVU ICD, BER $= 10^{-5}$.

A Additional Data

A.1 Frame Overhead Tables

This appendix provides the detailed frame overhead calculations used in the data budget. All values are in bytes.

A.1.1 Downlink Frame Types

Table 6: Detailed overhead of downlink frame types.

Frame type	AX.25	Header	Extra	Payload	Total
Full-size data frame	18	5	0	230	253
Picture chunk	18	5	1	64	88
Short telemetry (OBC)	18	5	7	14	44
Medium telemetry (EPS)	18	5	7	30	60
Transmitter telemetry	18	5	7	23	53
Receiver telemetry	18	5	7	14	44

The “Extra” column accounts for fields that are specific to certain frame types, such as the Picture ID for picture chunks or the telemetry preamble for telemetry frames.

A.1.2 Uplink Frame Types

Table 7: Detailed overhead of uplink frame types.

Frame type	AX.25	Header	Extra	Payload	Total
Full-size uplink frame	18	7	0	193	218
Code chunk	18	7	6	64	95
Picture uplink chunk	18	7	6	64	95

A.2 Step-by-Step Calculations

This section provides the step-by-step calculations for the three scenarios presented in Section 2.

A.2.1 Scenario 1: Telemetry-Only

$$R_{\text{eff}} = 9600 \cdot 0.50 = 4800\,\text{bit/s} = 600\,\text{bytes/s} \tag{21}$$
$$V_{\text{pass}} = 600 \cdot 600 \cdot 0.8 = 288{,}000\,\text{bytes} \approx 281\,\text{KB} \tag{22}$$
$$V_{\text{day}} = 281 \cdot 3 = 843\,\text{KB} \approx 0.84\,\text{MB} \tag{23}$$

A.2.2 Scenario 2: Payload-Only

$$R_{\text{eff}} = 9600 \cdot 0.727 = 6979\,\text{bit/s} \approx 872\,\text{bytes/s} \tag{24}$$
$$V_{\text{pass}} = 872 \cdot 600 \cdot 0.8 = 418{,}560\,\text{bytes} \approx 409\,\text{KB} \tag{25}$$
$$V_{\text{day}} = 409 \cdot 3 = 1227\,\text{KB} \approx 1.20\,\text{MB} \tag{26}$$

A.2.3 Scenario 3: Mixed

$$\eta_{\text{mixed}} = 0.25 \cdot 0.50 + 0.75 \cdot 0.727 = 0.670 \tag{27}$$
$$R_{\text{eff}} = 9600 \cdot 0.670 = 6432\,\text{bit/s} = 804\,\text{bytes/s} \tag{28}$$
$$V_{\text{pass}} = 804 \cdot 600 \cdot 0.8 = 385{,}920\,\text{bytes} \approx 377\,\text{KB} \tag{29}$$
$$V_{\text{day}} = 377 \cdot 3 = 1131\,\text{KB} \approx 1.10\,\text{MB} \tag{30}$$

A.3 Parameter Sources

Table 8: Source of each input parameter.

Parameter	Value	Source
Downlink frequency	430–450 MHz	TRXVU ICD, Table 4-1
Uplink frequency	140–150 MHz	TRXVU ICD, Table 4-2
Downlink modulation	BPSK	TRXVU ICD, Table 4-1
Uplink modulation	FSK	TRXVU ICD, Table 4-2
Downlink data rate	9600 bit/s	TRXVU ICD, Table 4-1
Uplink data rate	9600 bit/s	TRXVU ICD, Table 4-2
Downlink max payload	235 bytes	TRXVU ICD, Table 4-1
Uplink max payload	200 bytes	TRXVU ICD, Table 4-2
AX.25 metadata	18 bytes	Frame Design Manual, Sec. 1.1
Downlink header	5 bytes	Frame Design Manual, Table 3
Uplink header	7 bytes	Frame Design Manual, Table 1
Picture chunk data	64 bytes	Transmission Manual, Sec. 3.2.3
Picture ID	1 byte	Transmission Manual, Sec. 3.2.3
Telemetry preamble	7 bytes	Transmission Manual, Sec. 3.5.6.2

A.4 Packet Error Rate Calculation

The packet error rate (PER) is calculated from the bit error rate (BER) and the frame length L in bits:

$$\text{PER} = 1 - (1 - \text{BER})^{L}. \tag{31}$$

Using $\text{BER} = 10^{-5}$ as specified in the TRXVU ICD:

$$L = 235\,\text{bytes} = 1880\,\text{bits}: \quad \text{PER} \approx 1.6\% \tag{32}$$

$$L = 200\,\text{bytes} = 1600\,\text{bits}: \quad \text{PER} \approx 1.59\% \tag{33}$$

$$L = 88\,\text{bytes} = 704\,\text{bits}: \quad \text{PER} \approx 0.70\% \tag{34}$$

These values indicate that frame losses due to bit errors are small but not negligible. Retransmissions should be accounted for in the operational downlink strategy.